

- 3 (a) (i) work done equals force $\times$ distance moved / displacement in the direction of the force B1 [1]
- (ii) power is the rate of doing work / work done per unit time B1 [1]
- (b) (i) kinetic energy $= \frac{1}{2} mv^2$ C1
 $= 0.5 \times 600 (9.5)^2$ C1
 $= 27075 \text{ (J)} = 27 \text{ kJ}$ A1 [3]
- (ii) potential energy $= mgh$ M1
 $= 600 \times 9.81 \times 4.1$ A1
 $= 24132 \text{ (J)}$ A0 [2]
 $= 24 \text{ kJ}$
- (iii) work done $= 27 - 24 = 3.0 \text{ kJ}$ A1 [1]
- (iv) resistive force $= 3000 / 8.2$ (distance along slope $= 4.1 / \sin 30^\circ$) C1
 $= 366 \text{ N}$ A1 [2]

- 4 (a) clamped horizontal wire over pulley or vertical wire attached to ceiling with mass